

SIDDHI PRAVIN LIPARE

MS by Research · CVIT, IIIT Hyderabad

siddhi.l@research.iiit.ac.in | siddhi.lipare@gmail.com | [GitHub](#) | [LinkedIn](#)

EDUCATION

MS by Research, Computer Science & Engineering	Jan 2026 – Present
<i>International Institute of Information Technology, Hyderabad</i>	CGPA 8.00
BS Data Science & Engineering	Dec 2021 – May 2025
<i>Indian Institute of Science Education & Research, Bhopal</i>	CGPA 8.41
Senior Secondary (Class XII), CBSE Board · 90.0%	2021
Secondary (Class X), CBSE Board · 95.0%	2019

EXPERIENCE

Centre for Visual Information & Technology (CVIT), IIIT Hyderabad	May 2025 – Dec 2025
<i>Applied Research Fellow</i>	Offer Letter
- Designed RoadTones – a dataset–model–evaluation framework for tone-controllable road-video captioning. Work accepted at CVPR Findings 2026 (Project page). - Developed key components of the dataset generation pipeline and evaluation metrics, enabling comprehensive benchmarking against 10 state-of-the-art Video-Language Models (VLMs). - Investigated and optimized the vLLM inference pipeline for the project demo, minimizing video quality degradation during inference to preserve caption generation performance.	
Indian Institute of Technology Madras	May 2024 – Jul. 2024
<i>Research Intern</i>	Certificate , Report
- Explored Super-Resolution techniques for 3D medical imaging, enhancing z-axis resolution in CT and MRI scans. - Utilized GANs, Diffusion Models, and NeRF to improve image clarity for better diagnostics.	

PROJECTS

BS Thesis – Image Tampering Detector	Jan. 2025 – April 2025
<i>IISER Bhopal</i>	Report
Leveraged the use of Discrete Wavelet Transform along with an Adaptive Sub-Band Attention Module to capture frequency-spatial cues to enable precise localization of tampered regions in an image.	
Improving Monocular Depth Estimation of Outdoor Images in Dim Light Conditions	Sept. 2024 – Nov. 2024
<i>IISER Bhopal</i>	Report
Integrated denoising and deblurring filters into the transformer-based DepthAnything V2 model to improve depth estimation in blurry and noisy nighttime images.	
Prediction of Mortality Rate of Heart Failure Patients Admitted to ICU	Sept. 2023 – Nov. 2023
<i>IISER Bhopal</i>	GitHub
A binary classification problem executed using supervised machine-learning techniques. Several classifiers and pre-processing methods were applied to determine which classifier gave the best results.	

TECHNICAL SKILLS

Languages	Python3, C/C++ , JavaScript, MATLAB, R, HTML, CSS, MySQL
Technologies	PyTorch, TensorFlow, Keras, OpenCV
Other	Docker, Firebase, Android SDK, Data Structures & Algorithms, Linux

KEY COURSES

Mathematics	Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability & Statistics
DS & Engineering	3D Deep Learning, Natural Language Processing, Data Analyses & Statistics for Geosciences, Applied & Accelerated AI, Deep Learning, Artificial Intelligence, Machine Learning, Computer Vision, Database Management Systems, Applied Optimisation, Econometrics, Data Science in Practice (Advanced Python), Data Structures & Algorithms, Signals & Systems

ACHIEVEMENTS

Publication — ‘RoadTones: Tone Controllable Text Generation from Road Event Videos’ presented at CVPR Findings and WiCV@CVPR Workshop .	2026
Presentation — Poster ‘Improving Image Tampering Detection: A Hybrid Approach Using Frequency and Spatial Domain Features’ presented at CV4Science@CVPR Workshop .	2025
ISRO Robotics Challenge — Among the Top 30 teams out of 400 in India.	2024
IEEE CAS Student Design Competition — 3rd place in the Asia-Pacific Region.	2022